

National Engineering Degree

Academic year 2025/2026



Department to be adapted

Specialization to be adapted

Final-Year Engineering Project Report

Discipline: Data Science

Final-year engineering project title

Titre du projet de fin d'études

Prepared by	First name LAST NAME
Host organization	Host company / laboratory
Industrial supervisor	Name of industrial supervisor
Academic supervisor	Name of academic supervisor
Defense date	Month 2026
Confidentiality	Public
Jury chair	Name of jury chair
Examiner	Name of jury member
Examiner	Name of jury member

Writing and compliance guide

This page should remain formal, sober and institutionally consistent. The official school logo is already integrated in the template. If the project is confidential, prepare an academic version that preserves the scientific logic while masking sensitive elements.

Dedication

Writing and compliance guide

Optional section. Recommended length: half a page to one page. Keep it sober and appropriate for an academic document.

[To complete: Write a short personal dedication.]

Acknowledgements

Writing and compliance guide

Thank supervisors, the host company, the jury and contributors in a concise and professional way. Avoid overly long acknowledgements, excessive wording or non-academic tone.

[To complete: Write concise and professional acknowledgements.]

Abstract

Abstract

Writing and compliance guide

150 to 250 words. The abstract should be self-contained and cover: context, problem statement, methodology, main results and contribution. Add 3 to 6 keywords.

[To complete: Write the abstract in English.]

Keywords: keyword 1, keyword 2, keyword 3

Optional French summary

Writing and compliance guide

Add this subsection only if your department or jury requires a French summary. Keep the same scientific content and avoid literal translation errors.

[To complete: If required, write the French summary here.]

Mots-clés : mot-clé 1, mot-clé 2, mot-clé 3

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List of abbreviations and acronyms

Writing and compliance guide

Sort acronyms alphabetically and define each one the first time it appears in the report.

AI	Artificial Intelligence
API	Application Programming Interface
CAD	Computer-Aided Design
CNN	Convolutional Neural Network
FPGA	Field-Programmable Gate Array
IoT	Internet of Things
KPI	Key Performance Indicator
PFE	Final-Year Project
RMSE	Root Mean Square Error

Integrated institutional guide for the final-year project

Writing and compliance guide

This section turns the template into a practical reading guide for the final-year project framework. It is visible only when guide mode is activated. For the clean submission version, disable it by replacing `\showguidetrue` with `\showguidefalse` in `config.tex`.

1. Objectives of the final-year internship/project

The final-year internship/project is both a synthesis exercise and a professionalization experience. Its purpose is to allow the engineering student to demonstrate the ability to address a complex engineering problem, analyze its stakes, propose a rigorous approach, and develop and validate an argued solution in a real or realistic context. The report is therefore not a simple activity log, but an academic proof of competence. At minimum, the internship/project should enable the student to:

- analyze the context, problem statement, needs, constraints and scope of the topic;
- mobilize a critical state of the art, standards, tools, methods or existing solutions;
- justify technical, scientific, methodological or economic choices;
- design, implement, model, experiment with or integrate a solution;
- validate results through an explicit protocol and relevant indicators;

- demonstrate autonomy, professionalism, scientific integrity and clear communication skills.

Writing and compliance guide

In the general introduction, reformulate the topic as a problem statement and announce measurable objectives. In the conclusion, explicitly return to the degree of achievement of these objectives.

2. What the report must demonstrate

The reader must be able to reconstruct the following logical chain: **problem or need** → **requirements** → **method** → **proposed solution** → **validation** → **results** → **conclusion**. In practice, this means that the report should present in a traceable way:

- the context and positioning of the project;
- functional and non-functional needs, constraints and assumptions;
- design choices and their justifications;
- the actual implementation or experimentation;
- a test plan or validation protocol;
- a critical discussion: limitations, deviations, threats to validity and perspectives.

Writing and compliance guide

A strong final-year report is not only descriptive. It explains how and why decisions were made, and how their relevance was verified.

3. Recommended evaluation scheme

The evaluation of the final-year project should remain readable, homogeneous and aligned with the targeted competencies. The following table can remain in the template so that the student understands, from the writing phase onward, what will be observed by the jury.

Criterion	Observable indicators	Indicative weight
Quality of framing and analysis	Problem statement, objectives, needs, scope and initial positioning.	10%
State of the art and use of sources	Relevance, diversity, critical thinking, citation quality and project positioning.	10%
Methodology and rigor	Justified choices, structured approach, traceability, engineering logic.	15%
Design and implementation	Architecture, implementation, integration, technical quality of the solution.	20%
Validation and discussion	Protocols, results, critical analysis, comparison with objectives.	15%
Report quality	Structure, clarity, style, figures, tables and formal consistency.	10%
Oral defense	Clarity, mastery of the topic, time management, quality of presentation material.	10%
Answers to questions and critical distance	Argumentation, autonomy, lucidity regarding limitations and perspectives.	10%

Writing and compliance guide

Use this grid as a self-check tool. Before submission, make sure that each criterion is visible in the report or prepared for the oral defense.

4. Bibliographic references: requirements and good practices

The bibliography is one of the strongest indicators of the scientific quality of a report. Any borrowed idea, data, figure, table, code excerpt, formula or method must be properly attributed. The report should respect the following principles:

- use a consistent citation style; the **IEEE** style is recommended for engineering programs;
- ensure that every source cited in the text appears in the bibliography and vice versa;
- prioritize indexed journal papers, recognized conference proceedings, academic books, standards, patents and reliable technical reports;
- target at least **25 references** for a master/engineering degree level report, with $\geq 60\%$ indexed scientific sources whenever applicable;
- avoid Wikipedia, non-specialized blogs, unverifiable commercial content and websites without clearly identifiable authorship;
- add an access date for web resources.

4.1 How to cite in the text

In IEEE style, citations appear in numbered brackets such as [1], [2] or [3], [4]. The citation should appear as close as possible to the corresponding idea. A paraphrase must be cited just like a direct quotation.

Example: *Recent work shows that system robustness strongly depends on training-data quality and on the validation protocol [1].*

Example: *The selected architecture is consistent with practices described in technical literature [2, 3].*

4.2 Examples of IEEE references

- **Journal article:** [1] A. Author and B. Author, « Article title, » *Journal name*, vol. X, no. Y, pp. xx–yy, year.
- **Conference paper:** [2] A. Author, « Paper title, » in *Proc. Conference name*, city, country, year, pp. xx–yy.
- **Book:** [3] A. Author, *Book title*, Xth ed. city: publisher, year.
- **Standard:** [4] *Standard title*, organization, reference, year.
- **Website:** [5] Author or organization, « Resource title, » website, year. [Online]. Available: URL. [Accessed: day month year].

4.3 How to fill in `references.bib`

The `references.bib` file provided with this template allows students to manage their references properly with BibTeX. The minimum workflow is the following:

1. add a bibliographic entry to `references.bib`;
2. cite the source in the text using `\cite{key}`;
3. compile the project with the standard Overleaf sequence;
4. verify that all references are correct, complete and homogeneous.

4.4 BibTeX examples to copy and adapt

```
@article{myPaper2025 ,
  author   = {A. Author and B. Author},
  title    = {Article title},
  journal  = {Journal name},
  volume   = {12},
  number   = {3},
  pages    = {101--115},
  year     = {2025}
}

@inproceedings{myConf2024 ,
  author    = {A. Author and B. Author},
  title     = {Paper title},
  booktitle = {Proceedings of ...},
  pages     = {55--60},
  year      = {2024}
}

@book{myBook2023 ,
  author    = {A. Author},
  title     = {Book title},
  publisher = {Publisher},
  year      = {2023}
```

```

}

@misc{myWeb2026,
  author = {{Organization name}},
  title  = {Web resource title},
  year   = {2026},
  url    = {https://example.org},
  note   = {[Online]. Accessed: 15 April 2026}
}

```

Writing and compliance guide

A long bibliography does not automatically improve report quality. Juries value a selection that is relevant, reasonably recent, scientifically credible and actually used in the analysis.

5. How to use this template pedagogically

It is recommended that students read this section before starting to write, then come back to it at three key moments:

- at the beginning, to understand the academic purpose of the internship/project;
- midway through, to check the alignment between objectives, chapters and produced evidence;
- before submission, to verify the report's compliance with institutional expectations.

Writing and compliance guide

Practical suggestion for supervisors: keep this section in the working version shared with the student, then hide it in the final submission version. The template then serves a dual purpose: **report model** and **reading support for the final-year project framework**.

General introduction

Writing and compliance guide

This introduction should clearly present: the general context, the problem statement, the objectives, the scope, the host organization if relevant, and the structure of the report. Recommended length: 1 to 2 pages. Do not insert figures or tables in this part.

In a context of rapid digital and industrial transformation, final-year projects constitute a summative learning situation in which the engineering student must demonstrate the ability to analyze a complex problem, propose a rigorous resolution approach and validate a solution with clear scientific and technical arguments.

This work is carried out within the framework of a project conducted at Host company / laboratory. The topic focuses on **[To complete: state the topic in one clear sentence]**. The main problem statement may be formulated as follows: **[To complete: formulate the central project question]**. The selected objectives aim to **[To complete: list the specific objectives in coherence with the validated topic]**.

The project scope covers **[To complete: indicate what is included and what is outside the scope]**. This clarification is essential to avoid a purely descriptive report without real engineering framing. Mechatronics-oriented project: explain system decomposition, mechanical-electronic-control coupling, CAD aspects, kinematics/dynamics, integration strategy, performance testing and operational safety.

This report is organized as follows. The first chapter presents the state of the art and the project positioning. The second chapter describes the requirements analysis, specifications and methodology. The third chapter details the design and architecture.

The fourth chapter addresses implementation or experimentation. The fifth chapter presents testing, validation and critical discussion of the results. Finally, the general conclusion synthesizes the contributions of the work and outlines future perspectives.

CHAPTER 1

State of the art and project positioning

Writing and compliance guide

This chapter should demonstrate a critical review of the literature or existing solutions, not a simple compilation. Prioritize indexed papers, conferences, standards, patents and authoritative technical documentation. End the chapter with the explicit positioning of the project.

1.1 Scientific and technical context

[To complete: Introduce the field and the main stakes of the topic.]

1.2 Critical analysis of existing work

[To complete: Compare 3 to 5 relevant approaches, tools or architectures.]

Table 1.1 – Example of a comparative state-of-the-art table

Reference / solution	Principle / contribution	Identified limi- tations	Relevance to the project
Solution A	[To complete: ...]	[To complete: ...]	[To complete: ...]
Solution B	[To complete: ...]	[To complete: ...]	[To complete: ...]
Solution C	[To complete: ...]	[To complete: ...]	[To complete: ...]

1.3 Project positioning

[To complete: Show the uncovered need, the expected contribution and the justification of the selected approach.]

Writing and compliance guide

Strong expectations in an ASIIN/EUR-ACE-style reading: relevance of sources, quality of critical distance, and clear linkage between identified gaps and project choices.

CHAPTER 2

Requirements analysis, specifications and methodology

Writing and compliance guide

The reader must be able to reconstruct the chain need $\rightarrow$ requirements $\rightarrow$ method.

This chapter is essential to demonstrate mastery of engineering framing.

2.1 Analysis of the existing situation and usage context

[To complete: Present the real project environment, users, and organizational, technical or regulatory constraints.]

2.2 Functional and non-functional requirements

Table 2.1 – Example of requirements formalization

ID	Requirement / need	Type	Acceptance criterion
EX-01	[To complete: Measure / detect / predict ...]	Functional	[To complete: Expected accuracy]
EX-02	[To complete: Response time]	Non-functional	[To complete: < 200 ms]
EX-03	[To complete: Safety / robustness / reliability]	Constraint	[To complete: Validation condition]

2.3 Specifications and performance indicators

[To complete: Link requirements to metrics or future tests.]

2.4 Selected methodology

[To complete: Describe the approach: development cycle, experimental protocol, tools, milestones, datasets, test bench, design model, etc.]

2.4.1 Discipline-specific orientation

Writing and compliance guide

Mechatronics-oriented project: explain system decomposition, mechanical-electronic-control coupling, CAD aspects, kinematics/dynamics, integration strategy, performance testing and operational safety.

[To complete: Insert diagrams or models adapted to the discipline here: UML/SysML, functional scheme, network architecture, data-flow diagram, FAST, SADT, state diagrams, etc.]

CHAPTER 3

Design and architecture

Writing and compliance guide

This chapter must justify technical choices. Describing the architecture is not enough; you must explain why it makes sense with respect to the requirements and the usage context.

3.1 Overall solution architecture

[To complete: Present the global architecture with a clear diagram.]

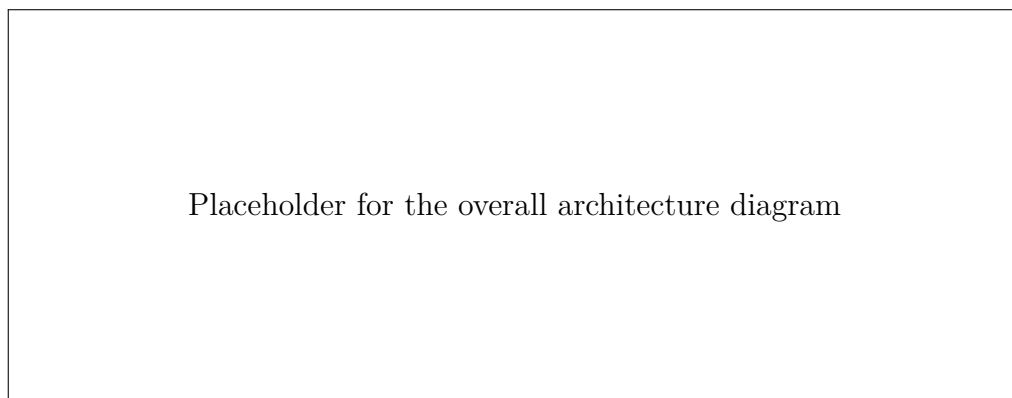


Figure 3.1 – Proposed overall architecture

3.2 Subsystem decomposition

[To complete: Describe the subsystems, their responsibilities and their interfaces.]

3.3 Argued comparison of alternatives

Table 3.1 – Example of a decision-support matrix

Criterion	Option 1	Option 2	Option 3	Justification of the choice
Cost	3	2	1	[To complete: ...]
Performance	2	3	1	[To complete: ...]
Ease of integration	3	2	2	[To complete: ...]
Robustness	2	3	2	[To complete: ...]

3.4 Requirements-to-design traceability

[To complete: Introduce a traceability table linking requirements, design choices and selected components.]

CHAPTER 4

Realization, implementation or experimentation

Writing and compliance guide

Document what has actually been achieved. The jury expects evidence of implementation, integration, encountered difficulties and adopted solutions. Avoid long screenshots or code blocks without analytical commentary.

4.1 Hardware and software environment

[To complete: Present platforms, libraries, software versions, sensors, boards, simulation tools, IDEs, frameworks or CAD software.]

4.2 Implementation steps

[To complete: Describe the main implemented functionalities, important algorithms and parameter settings.]

Algorithm 1 Example of pseudo-code for a core functionality

- 1: Initialize components / variables
 - 2: Acquire inputs or load data
 - 3: Apply the processing / model / control logic
 - 4: Produce an exploitable output or control action
 - 5: Log relevant results for validation
-

4.3 Difficulties encountered and solutions

[To complete: Show the student's capacity for analysis and adaptation.]

4.4 Produced technical deliverables

[To complete: Prototype, code, interface, schematics, PCB, mock-up, dashboard, model, enriched dataset, etc.]

CHAPTER 5

Testing, validation and discussion

Writing and compliance guide

This chapter is decisive. Results must be analyzed, not only displayed. The reader must understand what was tested, how, under which conditions, with which indicators and with what level of confidence.

5.1 Validation plan

[To complete: Describe the different test types: unit, integration, system, field, experimental validation, comparison with the state of the art, etc.]

5.2 Protocols and experimental conditions

[To complete: Describe datasets, test benches, sensors, scenarios, parameters and indicators.]

5.3 Presentation of results

Table 5.1 – Example of test summary

Test	Objective	Expected	Obtained	Conclusion
T1	[To complete: ...]	[To complete: ...]	[To complete: ...]	[To complete: Validated / to improve]
T2	[To complete: ...]	[To complete: ...]	[To complete: ...]	[To complete: Validated / to improve]
T3	[To complete: ...]	[To complete: ...]	[To complete: ...]	[To complete: Validated / to improve]

5.4 Critical discussion

[To complete: Interpret the results, explain deviations, discuss limitations, threats to validity and possible improvements.]

Writing and compliance guide

Mechatronics-oriented project: explain system decomposition, mechanical-electronic-control coupling, CAD aspects, kinematics/dynamics, integration strategy, performance testing and operational safety.

General conclusion and perspectives

Writing and compliance guide

The conclusion must explicitly answer the objectives announced in the introduction. Do not introduce new results. End with coherent, realistic and well-argued perspectives.

This project addressed the problem related to **[To complete: recall the topic]**. The work carried out led to **[To complete: summarize the main results and contributions]**. The objectives achieved include in particular **[To complete: list the main outcomes]**, while some limitations remain, especially **[To complete: indicate the technical, methodological or organizational limits]**.

From both an academic and professional perspective, this work highlights **[To complete: the mobilized competencies: analysis, design, validation, critical thinking, autonomy]**. The future perspectives concern **[To complete: future improvements, scope extension, industrialization, scale-up, new datasets, on-site validation, etc.]**.

Bibliography

- [1] A. Example and B. Example, “An example of engineering journal article citation,” *International Journal of Engineering Examples*, vol. 12, no. 3, pp. 101–115, 2024.
- [2] C. Example and D. Example, “An example of conference paper citation,” in *Proceedings of the International Conference on Applied Engineering*, 2023, pp. 55–60.
- [3] International Organization for Standardization, “Example standard for technical documentation,” 2022, iSO Example 12345:2022.

CHAPTER A

Recommended traceability tables

Writing and compliance guide

This appendix provides recommended structures to strengthen report coherence. Students may integrate all or part of them in the main body if it improves readability.

A.1 Requirements Traceability Matrix

Requirement	Design choice	Implemented element	Validation method	Status
EX-01	[To complete: ...]	[To complete: ...]	[To complete: ...]	Validated
EX-02	[To complete: ...]	[To complete: ...]	[To complete: ...]	In progress

A.2 Concept Comparison Matrix

Criterion	Idea 1	Idea 2	Idea 3	Comment / final choice
-----------	--------	--------	--------	------------------------

Performance	3	2	1	[To complete: ...]
Cost	2	3	1	[To complete: ...]
Risk	2	2	3	[To complete: ...]

A.3 Validation Summary Table

Test	Objective	Indicator	Expected	Obtained	Conclusion
T1	[To complete: ...]	[To complete: ...]	[To complete: ...]	[To complete: ...]	OK
T2	[To complete: ...]	[To complete: ...]	[To complete: ...]	[To complete: ...]	KO

CHAPTER B

Scientific integrity, generative AI and reproducibility

B.1 Declaration of generative AI usage

Writing and compliance guide

The use of AI tools to rephrase text, generate code, create figures, suggest outlines or summarize resources must be explicitly declared. The student remains fully responsible for the accuracy of the content and compliance with citation rules.

Example statement: *Generative AI tools were used at certain stages to support text rephrasing, suggest a document structure or propose code examples. All content was subsequently verified, corrected and validated by the author, who assumes full scientific and academic responsibility for the final work.*

B.2 Minimum reproducibility expectations

- Datasets: source, size, date, acquisition mode, restrictions.
- Tools: software versions, libraries, hardware environment, system configuration.
- Critical parameters: hyperparameters, thresholds, frequencies, dimensions, protocols.
- Validation elements: scripts, test notebooks, configuration captures, schematics,

bills of materials.

- Confidentiality: distinguish what is shareable, masked or anonymized.

B.3 Compliance reminders

- ✓ Every borrowed figure, table, code excerpt or idea must be attributed.
- ✓ Results must be interpreted critically.
- ● An excessively high similarity rate may compromise report acceptability.
- ● Fluent text is not scientific evidence if it is not linked to a method and verifiable results.

CHAPTER C

Final checklist before submission

C.1 Structure and coherence

- ✓ Front matter is complete.
- ✓ The introduction announces the problem statement, objectives and plan.
- ✓ Each chapter has a clear internal logic.
- ✓ The conclusion explicitly answers the objectives.

C.2 Scientific quality

- ✓ Choices are justified and compared.
- ✓ Results are validated through an explicit protocol.
- ✓ Limitations and threats to validity are discussed.
- ✓ The report contains at least 25 references, with a majority of indexed scientific sources.

C.3 Form and integrity

- ✓ Figures and tables are numbered and cited in the text.
- ✓ SI units are respected.
- ✓ The bibliography follows a homogeneous style (IEEE recommended).

- ✓ The target similarity rate remains $< 15\%$.
- ✓ Generative AI usage is declared if applicable.

C.4 Files to submit

- Final PDF of the report.
- LaTeX source files (if required by the program).
- Similarity report.
- Oral defense slides.
- Technical appendices or complementary deliverables if required.